

Requested Patient-Level Models of Memory Modulation

Patient-level linear models using the behavioral first-session summary file

Requested outcome: avg_dprime_diff. The behavioral file does not contain a column with that exact name; the available patient-level outcome is avg_stim_dprime_diff, which was used here.

Critical modeling point: this file has one row per patient (54 rows, 54 unique patients). Because the outcome is already collapsed to a single number per person, a subject random intercept is not identifiable. The correct analysis at this level is a patient-level linear model, not an MLM with (1 | subject).

Executive Summary

- Model 1 includes sex and stim trajectory only.
- Model 2 includes sex, stim trajectory, stim hemisphere, stim amplitude, and age.
- Model 3 includes sex, stim trajectory, Memory_Z, and IED frequency.
- Model 4 includes sex, stim trajectory, stim hemisphere, stim amplitude, age, Memory_Z, and IED frequency.
- Model 5 includes Memory_Z and IED frequency only.

Interpretive frame

• Model 6 includes sex, stim trajectory, Memory_Z, and IED frequency. These models do not explain the original trial-level ICC directly. Instead, they test whether the requested patient-level covariates are associated with between-subject differences in the already-collapsed memory modulation score.

- Model 7 includes stim amplitude, stim hemisphere, and age only.

Data Availability and Descriptives

Table 1
Variable availability in the behavioral summary file

Variable	Label	Non-missing	Missing	Percent available
avg_stim_dprim	Memory modulation (avg_stim_dprime_diff)	54	0	100.0%
sex	Sex (male vs female)	54	0	100.0%
stim_trajectory	Stim trajectory (STG vs MTG)	48	6	88.9%
age	Age (years)	54	0	100.0%
stim_hemispher	Stim hemisphere (R vs L)	54	0	100.0%
stim_DB	Stim amplitude (DB)	54	0	100.0%
IED_freq	IED frequency (Likert)	44	10	81.5%
Memory_Z	Pre-surgical baseline memory (Memory_Z)	31	23	57.4%

Note. Stim trajectory is available for 48/54 patients after excluding the 3 frontal-trajectory cases from modeling.
Memory_Z is available for 31/54 patients and is concentrated in the Original, Timing, and Duration cohorts.
IED_freq is available for 44/54 patients.

In the raw patient-level outcome, males showed higher avg_stim_dprime_diff than females. That unadjusted difference is useful descriptively, but the requested models test whether sex and stimulation trajectory **Table 2** for patient-level variation alone or together with the additional covariates.

Outcome descriptives by sex

Group	N	Mean	SD	Median	Min	Max
female	34	-0.039	0.290	-0.053	-0.526	0.840
male	20	0.179	0.271	0.230	-0.779	0.440

Note. Means are computed on avg_stim_dprime_diff. Positive values indicate greater memory performance on stimulated relative to non-stimulated trials.

Memory_Z subset composition: Original (n=14), Timing (n=12), Duration (n=5). Frontal trajectories were excluded from the models using stim_trajectory, so the trajectory term compares STG against MTG only.

Model 1

Patient-level model with sex and stimulation trajectory.

Table 3
Model 1 fit summary

N	Predictors	R2	Adj R2	F(df)	Model p
48	2	0.254	0.221	7.677 (2, 45)	.001

Note. Model 1 includes sex and stim trajectory.

Table 4
Model 1 coefficients

Term	B	SE	t	p	95% CI
Intercept	-0.081	0.047	-1.731	.090	[-0.176, 0.013]
Sex (male vs female)	0.222	0.076	2.928	.005	[0.069, 0.374]
Stim trajectory (STG vs MTG)	0.216	0.104	2.077	.044	[0.007, 0.425]

Note. Female and MTG are the reference groups for binary predictors.

Model 1 was significant overall, $F(2, 45) = 7.677$, $p = .001$, and accounted for 0.254 of the variance in avg_stim_dprime_diff. The sex coefficient was $B = 0.222$ (95% CI [0.069, 0.374]), $t = 2.928$, $p = .005$. The stim trajectory coefficient was $B = 0.216$ (95% CI [0.007, 0.425]), $t = 2.077$, $p = .044$.

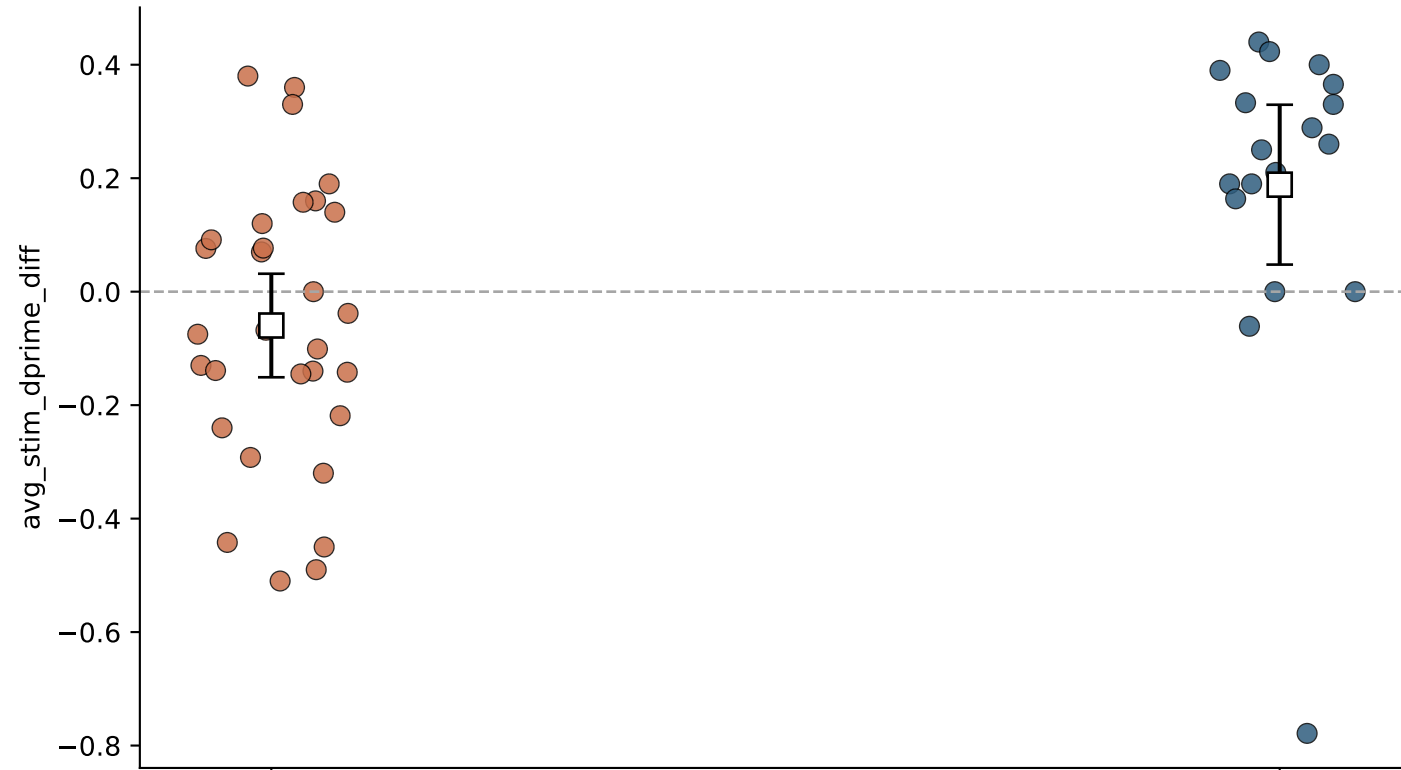
Table 5
Unique variance contribution in Model 1

Predictor	Delta R2	F change	p
Sex (male vs female)	0.142	8.571	.005
Stim trajectory (STG vs MTG)	0.071	4.314	.044

Note. Delta R2 values reflect the drop in model R2 when each predictor is removed from Model 1.

Unadjusted Sex Difference

Raw patient-level outcome by sex



Welch's test on the raw patient-level outcome gave $t(42.19) = 2.781$, $p = .008$, with males higher than females by 0.218 (95% CI [0.060, 0.377]). That unadjusted contrast is descriptive only and should be interpreted separately from the seven requested multivariable models.

Model Comparison

This page compares the seven patient-level models exactly as requested. Model 2 adds age, hemisphere, and stim amplitude to Model 1. Model 3 tests sex and trajectory together with Memory_Z and IED frequency. Model 4 includes all requested predictors in the complete-case subset. Model 5 isolates Memory_Z and IED frequency, Model 6 repeats sex plus trajectory with those same two covariates, and Model 7 isolates stim amplitude, stim hemisphere, and age.

Model	N	Predictors used	Omitted in subset	R2	Adj R2	Model p
Model 1	48	Sex (male vs female); Stim trajec		0.254	0.221	.001
Model 2	48	Sex (male vs female); Stim trajec		0.313	0.231	.006
Model 3	19	Sex (male vs female); Stim trajec		0.111	-0.143	.780
Model 4	19	Sex (male vs female); Stim trajec	Stim amplitude (DB) (no va	0.119	-0.322	.941
Model 5	22	Pre-surgical baseline memory (M		0.075	-0.022	.477
Model 6	19	Sex (male vs female); Stim trajec		0.111	-0.143	.780
Model 7	54	Stim amplitude (DB); Stim hemisphere (R vs L); Age (years)		0.147	0.096	.045

Note. Predictors with no variation inside a given subset are omitted automatically. Because Memory_Z and IED frequency are incomplete, Models 3 through 6 use smaller subsets than Models 1, 2, and 7.

Model 2

Patient-level model with sex, stimulation trajectory, stimulation hemisphere, stimulation amplitude, and age.

Table
Model 2 coefficients

Term	B	SE	t	p	95% CI
Intercept	0.087	0.184	0.470	.640	[-0.285, 0.458]
Sex (male vs female)	0.152	0.084	1.794	.080	[-0.019, 0.322]
Stim trajectory (STG vs MTG)	0.274	0.122	2.249	.030	[0.028, 0.521]
Stim hemisphere (R vs L)	-0.066	0.079	-0.836	.408	[-0.224, 0.093]
Stim amplitude (DB)	-0.270	0.190	-1.424	.162	[-0.652, 0.113]
Age (years)	0.002	0.004	0.434	.666	[-0.006, 0.009]

Note. Female, MTG trajectory, and left hemisphere are the reference groups when present. Constant predictors in a subset were omitted automatically.

N = 48, R2 = 0.313, adjusted R2 = 0.231, F(5, 42) = 3.823, p = .006. Omitted predictors: None.

Table
Model 2 unique variance contribution

Predictor	Delta R2	F change	p
Sex (male vs female)	0.053	3.220	.080
Stim trajectory (STG vs MTG)	0.083	5.058	.030
Stim hemisphere (R vs L)	0.011	0.698	.408
Stim amplitude (DB)	0.033	2.027	.162
Age (years)	0.003	0.188	.666

Note. Delta R2 values are specific to the analysis subset for that model.

Model 3

Patient-level model with sex, stimulation trajectory, baseline memory (Memory_Z), and IED frequency.

Table
Model 3 coefficients

Term	B	SE	t	p	95% CI
Intercept	0.023	0.119	0.192	.851	[-0.232, 0.278]
Sex (male vs female)	0.008	0.154	0.052	.959	[-0.323, 0.339]
Stim trajectory (STG vs MTG)	0.276	0.248	1.110	.286	[-0.257, 0.809]
Pre-surgical baseline memory (Memory_Z)	-0.026	0.107	-0.244	.811	[-0.255, 0.203]
IED frequency (Likert)	0.007	0.075	0.088	.931	[-0.153, 0.166]

Note. Female, MTG trajectory, and left hemisphere are the reference groups when present. Constant predictors in a subset were omitted automatically.

N = 19, R2 = 0.111, adjusted R2 = -0.143, F(4, 14) = 0.437, p = .780. Omitted predictors: None.
Table
Model 3 unique variance contribution

Predictor	Delta R2	F change	p
Sex (male vs female)	0.000	0.003	.959
Stim trajectory (STG vs MTG)	0.078	1.233	.286
Pre-surgical baseline memory (Memory_Z)	0.004	0.059	.811
IED frequency (Likert)	0.000	0.008	.931

Note. Delta R2 values are specific to the analysis subset for that model.

Model 4

Patient-level model including sex, stimulation trajectory, stimulation hemisphere, stimulation amplitude, age, baseline memory (Memory_Z), and IED frequency.

Table
Model 4 coefficients

Term	B	SE	t	p	95% CI
Intercept	0.038	0.261	0.145	.887	[-0.530, 0.606]
Sex (male vs female)	-0.000	0.168	-0.001	.999	[-0.367, 0.366]
Stim trajectory (STG vs MTG)	0.283	0.304	0.930	.371	[-0.379, 0.945]
Stim hemisphere (R vs L)	0.060	0.186	0.322	.753	[-0.346, 0.466]
Age (years)	-0.001	0.009	-0.113	.912	[-0.020, 0.018]
Pre-surgical baseline memory (Memo:	-0.018	0.134	-0.136	.894	[-0.310, 0.273]
IED frequency (Likert)	0.011	0.102	0.110	.914	[-0.210, 0.233]

Note. Female, MTG trajectory, and left hemisphere are the reference groups when present. Constant predictors in a subset were omitted automatically.

N = 19, R2 = 0.119, adjusted R2 = -0.322, F(6, 12) = 0.269, p = .941. Omitted predictors: Stim amplitude (DB) (no variation in analysis subset).

Table
Model 4 unique variance contribution

Predictor	Delta R2	F change	p
Sex (male vs female)	0.000	0.000	.999
Stim trajectory (STG vs MTG)	0.064	0.865	.371
Stim hemisphere (R vs L)	0.008	0.103	.753
Age (years)	0.001	0.013	.912
Pre-surgical baseline memory (Memory_Z)	0.001	0.018	.894
IED frequency (Likert)	0.001	0.012	.914

Note. Delta R2 values are specific to the analysis subset for that model.

Model 5

Patient-level model with baseline memory (Memory_Z) and IED frequency only.

Table
Model 5 coefficients

Term	B	SE	t	p	95% CI
Intercept	0.004	0.109	0.034	.973	[-0.224, 0.231]
Pre-surgical baseline memory (Memo:	-0.096	0.091	-1.053	.306	[-0.288, 0.095]
IED frequency (Likert)	0.058	0.068	0.863	.399	[-0.083, 0.200]

Note. Female, MTG trajectory, and left hemisphere are the reference groups when present. Constant predictors in a subset were omitted automatically.

N = 22, R2 = 0.075, adjusted R2 = -0.022, F(2, 19) = 0.771, p = .477. Omitted predictors: None.
Table
Model 5 unique variance contribution

Predictor	Delta R2	F change	p
Pre-surgical baseline memory (Memory_Z)	0.054	1.108	.306
IED frequency (Likert)	0.036	0.745	.399

Note. Delta R2 values are specific to the analysis subset for that model.

Model 6

Patient-level model with sex, stimulation trajectory, baseline memory (Memory_Z), and IED frequency.

Table
Model 6 coefficients

Term	B	SE	t	p	95% CI
Intercept	0.023	0.119	0.192	.851	[-0.232, 0.278]
Sex (male vs female)	0.008	0.154	0.052	.959	[-0.323, 0.339]
Stim trajectory (STG vs MTG)	0.276	0.248	1.110	.286	[-0.257, 0.809]
Pre-surgical baseline memory (Memory_Z)	-0.026	0.107	-0.244	.811	[-0.255, 0.203]
IED frequency (Likert)	0.007	0.075	0.088	.931	[-0.153, 0.166]

Note. Female, MTG trajectory, and left hemisphere are the reference groups when present. Constant predictors in a subset were omitted automatically.

N = 19, R2 = 0.111, adjusted R2 = -0.143, F(4, 14) = 0.437, p = .780. Omitted predictors: None.

Table
Model 6 unique variance contribution

Predictor	Delta R2	F change	p
Sex (male vs female)	0.000	0.003	.959
Stim trajectory (STG vs MTG)	0.078	1.233	.286
Pre-surgical baseline memory (Memory_Z)	0.004	0.059	.811
IED frequency (Likert)	0.000	0.008	.931

Note. Delta R2 values are specific to the analysis subset for that model.

Model 7

Patient-level model with stimulation amplitude, stimulation hemisphere, and age only.

Table
Model 7 coefficients

Term	B	SE	t	p	95% CI
Intercept	-0.041	0.172	-0.238	.813	[-0.387, 0.305]
Stim amplitude (DB)	-0.186	0.178	-1.048	.300	[-0.544, 0.171]
Stim hemisphere (R vs L)	-0.135	0.083	-1.630	.109	[-0.302, 0.031]
Age (years)	0.007	0.004	2.080	.043	[0.000, 0.015]

Note. Female, MTG trajectory, and left hemisphere are the reference groups when present. Constant predictors in a subset were omitted automatically.

N = 54, R2 = 0.147, adjusted R2 = 0.096, F(3, 50) = 2.872, p = .045. Omitted predictors: None.
Table
Model 7 unique variance contribution

Predictor	Delta R2	F change	p
Stim amplitude (DB)	0.019	1.098	.300
Stim hemisphere (R vs L)	0.045	2.657	.109
Age (years)	0.074	4.328	.043

Note. Delta R2 values are specific to the analysis subset for that model.

Bottom Line

This file supports a between-subject regression question, not a within-subject MLM question. The outcome is already one number per patient, so there is no residual subject-level nesting to model.

Model 1 (sex + trajectory) used $N = 48$ patients and gave $B = 0.222$ for sex ($p = .005$) and $B = 0.216$ for stim trajectory ($p = .044$).

Model 2 added stim hemisphere, stim amplitude, and age in the same trajectory-defined subset ($N = 48$; model $p = .006$).

Model 3 added Memory_Z and IED frequency instead ($N = 19$; model $p = .780$).

Model 4 included all requested predictors in the complete-case subset ($N = 19$; model $p = .941$). Model 5 tested only Memory_Z and IED frequency ($N = 22$; model $p = .477$). Model 6 tested sex, stim trajectory, Memory_Z, and IED frequency ($N = 19$; model $p = .780$). Model 7 tested stim amplitude, stim hemisphere, and age only ($N = 54$; model $p = .045$). If a predictor is constant within a subset, the script omits it automatically rather than estimating an uninterpretable coefficient.

Recommended interpretation

Interpret the seven models as separate requested patient-level regressions rather than as one stepwise sequence. Models 1 and 2 answer the trajectory-focused question in the larger subset, Models 3 and 6 place sex and trajectory alongside Memory_Z and IED frequency, Model 4 adds the full covariate set, Model 5 isolates Memory_Z plus IED frequency alone, and Model 7 isolates amplitude, hemisphere, and age alone.